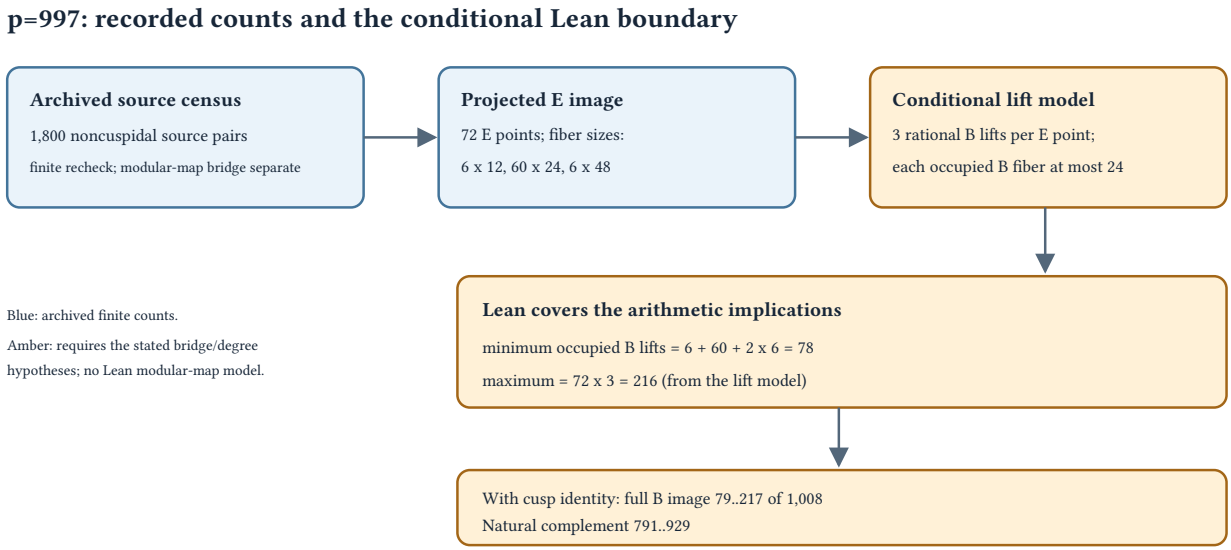
p=997 Hecke fiber proof and source review

Technical review T-1791, 2026-11-07. Reviewed source revision `927d1eada55ff51e87b89930cad48672147e0929`. The review plan was committed before dispatch at `639fb3de7ae6558a8beed6c119a820edefc266b1`.

Checked result

The blind reviewer independently derived all five natural-number statements and found exact counterexamples to three weakened controls. The source reviewer verified 17 of 17 seal entries, all 18 reviewed directory files against Git, and a byte-identical scratch audit replay. An independent finite-field reconstruction over all 997 j-values and ten special twists recovered 62 generic source classes and 1,800 paired chains; 2,306 archived edge occurrences were checked. Direct B-point enumeration confirmed both curve orders are 1,008, each saved three-point preimage coset is complete, and the archived E image has 72 points with fiber histogram 6 by 12, 60 by 24, and 6 by 48.

The archived audit still passed after changing a valid kernel root to an invalid one, changing a source j value, or swapping projections between two source rows. These controls delimit its consistency checks. They do not contradict the unmodified records, which passed the independent finite recheck. A full independent Sage quarter-shift reevaluation remains open.



Proof scope

The Lean declarations state conditional arithmetic. The full B-image interval 79 to 217 and complement 791 to 929 additionally use prior degree-24 modular-map, reduction, and cusp-image arguments. These geometric dependencies were not freshly proved in this packet. Explicit compatible assignments use either 78 or 216 noncuspidal B points while respecting the saved projected counts and cap; the histogram does not determine an exact B-image count.

The five arithmetic declarations have fresh machine status passed and faithful scoped fidelity. The concrete geometric image claim is partial in this packet, with review outcome inconclusive until its prior bridges are independently replayed. The archived audit's wider validation reading needs revision. No official hypothesis or evidence status changes here.

The archived build log pins Lean v4.35.0-rc4 and Mathlib 021c46bdd042f67b538b9f9871d556a93713a9a9 and lists only standard axioms. A fresh hosted pinned corpus run passed at review branch head 639fb3de7: GitHub Actions run 42339546067. The gate checked 32 added modules, including this p=997 source. The exact toolchain is absent locally.

Provenance and independent review

The prime screen was committed before the protocol's stated freeze. The protocol and resulting census/image first entered Git together afterward, so pre-execution protocol timing is declared rather than independently authenticated by Git. The reviewers filed separate attestations and did not read each other's reports. The existing independence checker returned exit 2 because the precommitted plan's `review_plan` field is an ID rather than the nested block its parser expects. The original plan and failure are preserved instead of retroactively replacing that prior. The diagram changes no canonical quantitative graph or dashboard input.

Next checks

Recompute every source-to-quarter-shift projection in Sage. Independently review the earlier degree, reduction, and cusp-image arguments. Strengthen the source audit so the three mutations fail, then rerun the affected source-to-claim review.

Full per-claim findings, code, controls, exact receipts and source-access lists are in the adjacent `report.md`, `blind/`, and `source/` artifacts.